

Cell	n	Med	High	f_{Med}	f_{High}
<i>BM Black male NR</i>	1168	198	312	.170	.267
<i>BF Black female NR</i>	346	56	75	.162	.217
<i>WM White male NR</i>	969	130	62	.134	.064
<i>WF White female NR</i>	312	70	20	.224	.064
<i>Black NR pooled</i>	1514	254	387	.168	.256
<i>White NR pooled</i>	1281	200	82	.156	.064

Table 1

Cell counts on the published ProPublica filter. Sex-marginal totals match `COMPASFromData.lean`; `score_text`-bucket splits are illustrative and sum to the published flagged totals (510, 131, 192, 90). Replace with extracted CSV counts in the Lean source if needed.

Appendix: A Deep ProPublica Certificate in TPTND-Lean

Abstract

This appendix accompanies the artifact at <https://github.com/dasaro/tptnd-lean>. It exhibits a fully compositional ProPublica COMPAS certificate that pulls every modelling decision the original Python pipeline performs implicitly — the `score_text` bucket collapse, the sex-stratified pooling, the conditioning on non-recidivists — into typed inference steps that `checkDerivation` verifies. The corresponding executable is `./lake/build/bin/pp_deep`; the source is `TPTND/PPDeep.lean`. Notational conventions in this appendix are deliberately compact: subscripts are dropped where unambiguous and rule presentations are abbreviated; the Lean source is the authoritative form.

1. Notation

We write $\sigma_{\rho s}$ for the provenance of the (race ρ , sex s) non-recidivist sub-cohort on the published 6172-row ProPublica filter, $\rho \in \{B, W\}$, $s \in \{M, F\}$. A frequency sequent $\Gamma \vdash_{\sigma} t_n : \alpha_f$ is rendered $-t : \alpha_f$ when context and provenance are clear from siblings. Output atoms used here: $\overset{n}{\text{Med}}$ (`score_text=Medium`), High (`score_text=High`), $\text{Flag} := \text{Med} + \text{High}$, NR (`two_year_recid=0`). We write $\mathcal{Q}$ for the two-sample score-test acceptance band on a difference of proportions; details in the main paper.

2. Compact rule presentation

For self-containment, the rules used in this appendix are recalled in Figure 1.

3. The deep certificate, top-down

Figure 2 shows the main spine of the certificate. The *Black side* is fully expanded; the *White side* is structurally identical, with the counts shown in Table 1 and provenance σ_{WM} , σ_{WF} substituted throughout. Five inference layers separate every leaf from the IUT2 root; EUT carries the gap interval into the typing context for downstream composition.

$$\begin{array}{c}
\frac{\sigma \neq \emptyset \quad x:\alpha_c \in \Gamma \quad nf \in \mathbb{N}}{\Gamma \vdash_{\sigma} t_n : \alpha_f} \text{OBS} \qquad \frac{\sigma_1 \cap \sigma_2 = \emptyset \quad \Gamma \vdash_{\sigma_1} t_n : \alpha_f \quad \Gamma \vdash_{\sigma_2} t_m : \alpha_g}{\Gamma \vdash_{\sigma_1 \uplus \sigma_2} t_{n+m} : \alpha_{(nf+mg)/(n+m)}} \text{UPDATE} \\[10pt]
\frac{\alpha \perp_{\text{syn}} \beta \quad \Gamma \vdash_{\sigma} t_n : \alpha_f \quad \Gamma \vdash_{\sigma} t_n : \beta_g}{\Gamma \vdash_{\sigma} t_n : (\alpha + \beta)_{f+g}} \text{I+} \qquad \frac{\Gamma \vdash_{\sigma} t_n : \alpha_f \quad \Delta \vdash_{\tau} u_m : \alpha_g \quad \sigma \cap \tau = \emptyset \quad \mathcal{Q}(n, m, f, g) = [\ell, h] \quad 0 \notin [\ell, h]}{\Gamma, \Delta \vdash \mathbf{UTrust}_{\mathcal{Q}}(t_n : \alpha_f; g, [\ell, h])} \text{IUT2} \\[10pt]
\frac{\Gamma \vdash \mathbf{UTrust}_{\mathcal{Q}}(t_n : \alpha_f; g, [\ell, h])}{\Gamma, x:\alpha_{\neg[\ell, h]} \vdash_{\sigma} t_n : \alpha_f} \text{EUT} \qquad \frac{x_i : \alpha_{a_i} \in \Gamma \ (i=1, \dots, k)}{\Gamma \vdash \Pi^{(\vec{a}, \vec{b})} : \alpha_p} \text{I-P} \qquad \frac{\Gamma \vdash \Pi^{(\vec{a}, \vec{b})} : \alpha_p \quad \text{observation } s/n}{\Gamma \vdash_{\Pi} H_j : \alpha_{P(H_j|d)}} \text{E-P}
\end{array}$$

Figure 1: Rules used in the deep derivation, in compact form. OBS requires a unique support entry $x:\alpha_c$ in Γ with x matching t 's atom name; I+ requires syntactic disjointness $\alpha \perp_{\text{syn}} \beta$; UPDATE requires disjoint provenances and a shared support entry; IUT2's acceptance band is computed by the Lean kernel and threaded through both premises; EUT carries the gap interval into the typing context as a transferable assumption; I-P/E-P introduce a parametric family of priors and produce a posterior under hypothesis H_j . Side conditions checked by `checkDerivation` are listed in the corresponding TPTND/Rules/*.lean files.

Black side (fully expanded).

$$\frac{\frac{\frac{\sigma_{BM} \vdash u_{1168}:\text{Med}_{198/1168} \text{ OBS} \quad \sigma_{BM} \vdash u_{1168}:\text{High}_{312/1168} \text{ OBS}}{\sigma_{BM} \vdash u_{1168}:\text{Flag}_{510/1168}} \text{ I+} \quad \frac{\sigma_{BF} \vdash u_{346}:\text{Med}_{56/346} \text{ OBS} \quad \sigma_{BF} \vdash u_{346}:\text{High}_{75/346} \text{ OBS}}{\sigma_{BF} \vdash u_{346}:\text{Flag}_{131/346}} \text{ I+}}{\sigma_B \vdash u_{1514}:\text{Flag}_{641/1514}} \text{ UPDATE}$$

White side (mirror, abbreviated).

$$\frac{\sigma_{WM} \vdash u_{969}:\text{Flag}_{192/969} \text{ (via I+ on Med, High)} \quad \sigma_{WF} \vdash u_{312}:\text{Flag}_{90/312} \text{ (via I+ on Med, High)}}{\sigma_W \vdash u_{1281}:\text{Flag}_{282/1281}} \text{ UPDATE}$$

Headline two-sample test.

$$\frac{\sigma_B \vdash u_{1514}:\text{Flag}_{641/1514} \quad \sigma_W \vdash u_{1281}:\text{Flag}_{282/1281} \quad \sigma_B \cap \sigma_W = \emptyset \quad \mathcal{Q}(1514, 1281, \frac{641}{1514}, \frac{282}{1281}) = [0.168, 0.238] \quad 0 \notin [0.168, 0.238]}{\sigma_B, \sigma_W \vdash \mathbf{UTrust}_{\mathcal{Q}}(u_{1514}:\text{Flag}_{641/1514}; \frac{282}{1281}, [0.168, 0.238])} \text{IUT2}$$

Extraction.

$$\frac{\sigma_B, \sigma_W \vdash \mathbf{UTrust}_{\mathcal{Q}}(\dots; \frac{282}{1281}, [0.168, 0.238])}{\sigma_B, \sigma_W, x_F:\text{Flag}_{\neg[0.168, 0.238]} \vdash_{\sigma_B} u_{1514}:\text{Flag}_{641/1514}} \text{EUT}$$

Figure 2: The deep ProPublica certificate, top-down. Five inference layers separate the deepest OBS leaf from the IUT2 verdict; the additional EUT step promotes the gap interval to a typing-context assumption that composes with downstream certificates. Output subscripts and context decorations are abbreviated; the full sequents appear in TPTND/PPDeep.lean.

4. Mapping the rules to ProPublica’s procedure

The published Larson–Mattu–Kirchner–Angwin methodology¹ performs, in order: (i) apply the 6172-row filter; (ii) condition on `two_year_recid=0` (non-recidivists); (iii) collapse `score_text` ∈ {Medium, High} into a single flagged event; (iv) pool sex within each race; (v) compare flagged rates across races. Steps (i) and (ii) live in the Lean leaves’ provenance discipline. The remaining three steps map onto rules of the calculus as follows.

ProPublica step	Rule used	What the kernel checks
(iii) <i>bucket collapse</i>	$4 \times \text{I}+$	$\text{Med} \perp_{\text{syn}} \text{High}$; same provenance and sample size on both observations; conclusion frequency $f_{\text{Med}} + f_{\text{High}}$
(iv) <i>sex pool</i>	$2 \times \text{UPDATE}$	$\sigma_M \cap \sigma_F = \emptyset$; conclusion frequency matches $(n_M f_M + n_F f_F) / (n_M + n_F)$
(v) <i>race comparison</i>	IUT2	$\sigma_B \cap \sigma_W = \emptyset$; the score-test band $\mathcal{Q}(1514, 1281, \frac{641}{1514}, \frac{282}{1281})$ is computed in exact rational arithmetic and 0 is verified to lie outside it
(post) <i>interval-as-context</i>	EUT	the resulting term claim’s typing context contains the complement-interval entry $x_F:\text{Flag}_{\neg[\ell, h]}$ and inherits the rest of the IUT2 conclusion’s context

5. Bayesian sister sub-tree

The Lean source also exhibits a separate I-P + E-P chain that posits three competing hypotheses for the Black FPR, $H_1:0.20$, $H_2:0.35$, $H_3:0.50$, with prior weights $b_i=a_i$ (rescaled to sum to one), and computes the posterior $P(H_j \mid d)$ given the observed 641/1514. Structurally this mirrors paper §4.1’s Bayesian audit and exercises the introduction and elimination of parametric prior families. It is a sister tree to the main spine, not part of it; the Lean code prints both verdicts side by side.

6. What this certificate does *not* include

The product rules $\text{I} \times / \text{E} \times$ and the arrow rules $\text{I} \rightarrow / \text{E} \rightarrow$ are implemented in the TPTND-Lean kernel but are not exercised by this certificate. An $\text{I} \rightarrow$ step would let us re-package the level-5 conclusion as a typed conditional certificate $(\text{NR} \Rightarrow_1 \text{Flag})_{641/1514}$, making the non-recidivist conditioning into an explicit typed step rather than a feature of the leaves’ provenance discipline; this is the direction predictive-parity-style audits (Northpointe’s counterclaim) take and is sketched in the main paper’s discussion of compositional rule use beyond marginal-event audits. The calculus admits both encodings; the present certificate uses the shallower one because the ProPublica narrative does too.

7. Reading the Lean source

The certificate above corresponds, line for line, to the `deepProPublicaCertificate` function in `TPTND/PPDeep.lean`. The single call to `checkDerivation` at the bottom of the function returns `Except.ok ()` when every side condition listed in Figure 1 is satisfied; the executable `./lake/build/bin/pp_deep` runs both the deep certificate and the Bayesian sister tree and exits non-zero on any failure.

¹<https://github.com/propublica/compas-analysis>